

1.a).

Let F be a secure PRP with block length 2. Consider the following encryption mode:

$\text{Enc}(k, m_i || \dots || m_\ell)$:

$\text{Co}\{0,1\}$

for $i = 1$ to ℓ :

$c_i := F(k, m_i) \text{ Ci-1 return coll...} || c_\ell$

Let F be a secure PRP with block length 2. Consider the following encryption mode: $\text{Enc}(k, :)(m;)$:
 $\text{Co}\{0,1\}$ for $i = 1$ to ℓ : return coll... || c_ℓ Write the corresponding decryption algorithm

In this problem, we need to design a secure PRP with block length 2. Let F be a secure PRP with block length 2, where $F(k, m_i) = eK m_i \text{ mod } p$. The algorithm will generate ciphertexts $c_{i-1} \dots c_\ell$ (where the i runs from 1 to ℓ) depending on the input r_i . It determines which of those two ciphertexts is more significant by tracing back through the MFS of the input r_i and $eK m_i \text{ mod } p$ in an appropriate order until a collision is found.

1.b).

A secure PRP is a secure polynomial PRP (defined in the paper). The presented mode achieves the security level of the original one, however, it is not CPA\$ secure. This is because an attacker can modify how their message will be encrypted (e.g., change its length).

Explanation:

Let F be a secure PRP with block length 2. Consider the following encryption mode: $\text{Enc}(k, m_i || \dots || m_\ell)$:
 $\text{Co}\{0,1\}$ for $i = 1$ to ℓ : $c_i := F(k, m_i) \text{ Ci-1 return coll...} || c_\ell$ where C_v is defined as follows: $C_v = F(k, m_i)$
 $v = \text{xor}(C_h) \text{ xor}(C_i)$; we assume that xor is 32bit random permutation and C_h is 128 bit random permutation. If encryption key K is used as seed we obtain: " $c_i = K[m_i] \text{ mod } 1$ & $c^\wedge = H[m_i] \text{ mod } 1$ & $c^\wedge = H[m_i+1] \text{ mod } 1$." Also H and v are independent thus proving that this mode does not achieve CPA\$ security

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2)

The given encryption scheme is vulnerable to a chosen plaintext attack (CPA) because an attacker can choose two plaintext messages m and m' and obtain their corresponding ciphertexts c and c' . The attacker can then compute the XOR of c and c' to obtain the XOR of m and m' (since XOR is linear), revealing information about the plaintext messages.

More formally, suppose an attacker selects two plaintext messages $m = m_1 || \dots || m'$ and $m' = m'_1 || \dots || m'$, where m and m' differ in at least one position. The attacker obtains the corresponding ciphertexts $c = \text{Enc}(k, m)$ and $c' = \text{Enc}(k, m')$. The attacker can then compute $c_{\text{xor}} = c \oplus c' = \text{Enc}(k, m) \oplus \text{Enc}(k, m') = \text{Enc}(k, m \oplus m')$.

The attacker has successfully obtained the encryption of the XOR of the two plaintext messages without knowing the key, which violates the indistinguishability requirement of CPA-security. Therefore, this encryption scheme is not CPA-secure.